How Europe's gas market reform changed the shape of its vulnerability

Thesis: gas market liberalisation reduced supplier power over volumes but created a new and more deniable channel of leverage, over the marginal price.



Pietro Mischi – Energy and Geopolitics

Two regimes of price formation

1962–2000s · **GRONINGEN MODEL**

PRICE FIXED BY CONTRACT

- 20–25 year contracts; oil-indexed formulas
- Take-or-pay: buyer committed to volumes
- Interruption = visible, costly, actionable breach
- Soviet gas continued to flow during the invasion of Afghanistan in 1979 and martial law in Poland in 1981

2003–today · **TTF hub**

PRICE SET AT THE MARGIN

- EU gas directives 1998 / 2003 / 2009 unbundle supply
- TTF launched 2003; hub indexation spreads in the 2010s
- The last unit traded prices all the gas
- Short-run demand is inelastic → small deficit, large repricing

Sources: Högselius (2013); EU gas directives 98/30/EC, 2003/55/EC, 2009/73/EC

Winter 2021

No valve was closed

Summer 2021 Rehden, Germany's largest storage site and run by a Gazprom subsidiary, is left almost empty.

Autumn 2021 Gazprom honours its contracts — but offers nothing extra on the spot market. The IEA says Russia could send more.

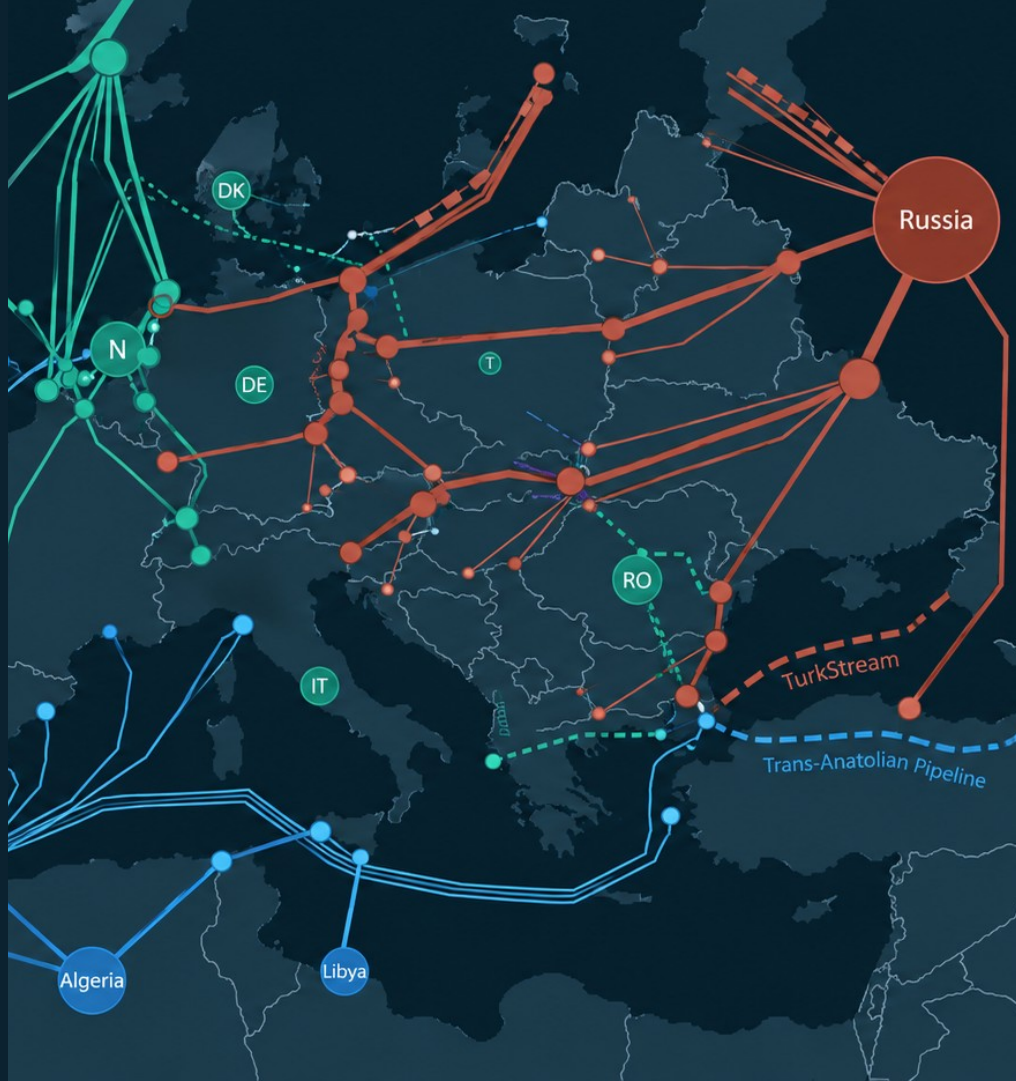
August 2022 The TTF price touches about €340/MWh.

<€20 → €340

per MWh — before the crisis vs. its peak

No breach, No casus belli, No single act to answer.

Withholding the marginal molecule repriced all the gas: [a chokepoint effect without holding the chokepoint](#) (Farrell & Newman 2019).



Europe's answer

BRUSSELS, 2022

For the first time, the fix is market design

Binding storage targets (Reg. 2022/1032)

and a cap on the TTF price (Reg. 2022/2578)

— an implicit admission that the market itself had become the vulnerability.

Liquidity cut both ways

The same hub pulled LNG from Asia within months — unthinkable in the Groningen era.

Intent stays unproven

Cold spring, weak wind, post-Covid demand. Conduct consistent with withholding, not proof.

The valve never left

2006 and 2009 show physical coercion survived. What changed was method and deniability.

1968 asked: from whom do we buy? 2021 asked: who sets the last price?

The pipeline decides whether gas arrives.

The design of the market decides what it can be made to cost.